# Xinlin Zhuang

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## Education

**East China Normal University**, ME in Computer Science and Technology — Sept 2023 – Jun 2026

- GPA: 3.71/4.0
- Honors: **ACL 2025 Best Theme Paper (3/3000+)**, University Outstanding Student Leader
- Awards: Fenzhong Social Scholarship, Kuanrui Corporate Scholarship
- Main Courses: Advanced Engineering Mathematics, Natural Language Processing, Academic Writing

**Ocean University of China**, BE in Computer Science and Technology — Sept 2019 – Jun 2023

- GPA: 3.80/4.0 Rank: 8/269
- Honors: Honorary Bachelor Degree **(Top 0.5%)**, Shandong Province Outstanding Graduate **(Top 1%)**, University Outstanding Student x 3
- Awards: National Scholarship, First-class University Scholarship x 3, AEON Corporate Scholarship
- Main Courses: Deep Learning, Linear Algebra, Probability Theory

## Experience

**Visiting Student**, MBZUAI – Abu Dhabi, UAE — May 2025 – Present

- Research goal: exploring machine learning methods on data processing for efficient large multimodal models training and meidical reasoning tasks.

**Algorithm Intern**, Bigdata Research Center, Shanghai AI Laboratory – China — Apr 2024 – Apr 2025

- Research goal: exploring machine learning methods on data processing (including data cleansing and data selection) for efficient large language models (LLMs) training.

## Projects

**Data Selection for Medical Text Reasoning** — May 2025 - Jul 2025

- Proposed a novel data selection strategy that combines medical difficulty score with influence score. This method enables models fine-tuned on **only 1%** of selected data to match full-dataset performance, while using **10%** consistently outperforms the random baseline. This work was submitted to AAAI 2026.
- Tools Used: Python, vLLM, Llama-Factory.

**Data Selection for Pre-training LLMs** — Apr 2024 - Apr 2025

- Proposed a regression based data selection framework to approximate validation loss on downstream tasks, combining a total of 25 data quality metrics, including natural quality signals, data importance scores, and model-based raters, for pre-training LLMs. The effectiveness was verified in extensive experiments with LLMs of 1B, 3B, and 7B scales. This work won ACL 2025 Best Theme Paper (3/3000+).
- Proposed a topic-based data mixing methodology for LLM pre-training, using a novel large-scale clustering approach to effectively categorize 600M documents. The effectiveness was verified in extensive experiments with LLMs of 1B, 3B, and 7B scales. This work was submitted to AAAI 2026.
- Explored developing lightweight BERT-style classifiers through distillation from LLM for extracting clean main content from raw HTML pages (including forum, shopping, and article pages).
- Tools Used: Python, Apache Spark, LM-Evaluation-Harness.

**Automatic Essay Comment Generation for Intelligent Education** — Dec 2023 - Feb 2024

- Proposed the first benchmark focusing on evaluating topic relevance of Chinese primary and middle school students essays (topic relevance level identification and comment generation). Experiments revealed that the combination of SFT and DPO significantly reduces hallucination of open-source LLMs on evaluating topic relevance and generating corresponding comments. This work was accepted at ACL 2024 Findings.
- Tools Used: Python, LabelStudio, Llama-Factory.

## Publications

**Meta-rater: A Multi-dimensional Data Selection Method for Pre-training Language Models** — **ACL 2025 Best Theme Paper (3/3000+)**

***Xinlin Zhuang***, Jiahui Peng, Ren Ma, YinFan Wang, Tianyi Bai, Xingjian Wei, Jiantao Qiu, Conghui He

**TOREE: Evaluating Topic Relevance of Student Essays for Chinese Primary and Middle School Education** — **ACL 2024 Findings**

***Xinlin Zhuang***, Hongyi Wu, Xinshu Shen, Peimin Yu, Gaowei Yi, Xinhao Chen, Tu Hu, Yang Chen, Yupei Ren, Yadong Zhang, Youqi Song, Binxuan Liu, Man Lan

**Harnessing Diversity for Important Data Selection in Pretraining Large Language Models** — **ICLR 2025 Spotlight (Top 5%)**

Chi Zhang, Huaping Zhong, Kuan Zhang, Chengliang Chai, Rui Wang, ***Xinlin Zhuang***, Tianyi Bai, Jiantao Qiu, Conghui He

**Towards Efficient Medical Reasoning with Minimal Fine-Tuning Data** — Submitted to AAAI 2026

***Xinlin Zhuang***, Feilong Tang, Haolin Yang, Shixiang Tang, Junjun He, Zongyuan Ge, Ying Qian, Imran Razzak

**Topic Over Source: The Key to Effective Data Mixing for Language Models Pre-training** — Submitted to AAAI 2026

Jiahui Peng*, ***Xinlin Zhuang****, Ren Ma, Jiantao Qiu, Conghui He

**Efficient Pretraining Data Selection for Language Models via Multi-Actor Collaboration** — **ACL 2025 Main**

Tianyi Bai, Ling Yang, Zhen Hao Wong, Jiahui Peng, ***Xinlin Zhuang***, Chi Zhang, Lijun Wu, Jiantao Qiu, Wentao Zhang, Binhang Yuan, Conghui He

**FinDABench: Benchmarking Financial Data Analysis Ability of Large Language Models** — **COLING 2025**

Shu Liu*, Shangqing Zhao*, Chenghao Jia*, ***Xinlin Zhuang****, Zhaoguang Long, Jie Zhou, Aimin Zhou, Man Lan

**Towards Explainable Chinese Native Learner Essay Fluency Assessment: Dataset, Tasks, and Method** — **EMNLP 2024 Findings**

Xinshu Shen, Hongyi Wu, Yadong Zhang, Man Lan, Xiaopeng Bai, Shaoguang Mao, Yuanbin Wu, ***Xinlin Zhuang***, Li Cai

**Are U a Joke Master? Pun Generation via Multi-Stage Curriculum Learning towards a Humor LLM** — **ACL 2024 Findings**

Yang Chen, Chong Yang, Tu Hu, Xinhao Chen, Man Lan, Li Cai, ***Xinlin Zhuang***, Xuan Lin, Xin Lu, Aimin Zhou

## Skills

**Programming Languages:** Python, C++

**Language:** CET-6 595, Preparing for IELTS

**Academic Service:** Active reviewer for conferences including *ACL ARR* and relative workshops, *IJCNN*, and journals including *Applied Intelligence*.

## Self-Evaluation

- I am passionate about exploring data-centric machine learning methods for boosting performance of LLMs and VLMs. Recently, I have been exploring data selection methods for pre-training VLMs.
- I am able to read and write academic papers fluently and communicate effectively in both daily and academic scenarios in English.